

Object Oriented Programming Lab – All Java Programs

1. Room Class

```
class Room {
    int no; String type; double area; boolean ac;
    void set(int n,String t,double a,boolean c){ no=n; type=t; area=a; ac=c; }
    void show(){ System.out.println(no+" "+type+" "+area+" "+ac); }
    public static void main(String[] args){
        Room r=new Room();
        r.set(101,"Deluxe",200.5,true);
        r.show();
    }
}
```

2(i). Static Variables, Methods & Blocks

```
class Test {
    static int x=10;
    static { System.out.println("Static Block"); }
    static void fun(){ System.out.println("Static Method "+x); }
    public static void main(String[] a){ fun(); }
}
```

2(ii). Greatest of 3 Numbers

```
class G {
    public static void main(String[] a){
        int x=10,y=40,z=20;
        System.out.println("Greatest = "+ (x>y?(x>z?x:z):(y>z?y:z)));
    }
}
```

3(i). Method Overriding

```
class A{ void show(){ System.out.println("A"); } }
class B extends A{ void show(){ System.out.println("B"); } }
class M{ public static void main(String[] a){ new B().show(); } }
```

3(ii). super Keyword

```
class A{ A(){ System.out.println("A"); } }
class B extends A{ B(){ super(); System.out.println("B"); } }
class M{ public static void main(String[] a){ new B(); } }
```

4. Polymorphism – Shape

```
class Shape{ void draw(){} void erase(){} }
class Circle extends Shape{ void draw(){System.out.println("Circle");}}
class Tri extends Shape{ void draw(){System.out.println("Triangle");}}
class Sq extends Shape{ void draw(){System.out.println("Square");}}
class M{ public static void main(String[] a){ Shape s=new Circle(); s.draw(); } }
```

5. Outer & Inner Class

```
class Outer {
    void show(){ System.out.println("Outer"); }
    class Inner { void show(){ System.out.println("Inner"); } }
    public static void main(String[] a){
        Outer o=new Outer(); o.show();
        Outer.Inner i=o.new Inner(); i.show();
    }
}
```

6. box & box3d

```
class Box{ int l,b; Box(int x,int y){l=x;b=y;} int area(){return l*b;} }
class Box3D extends Box{ int h; Box3D(int x,int y,int z){super(x,y);h=z;} int vol(){return l*b*h;} }
class M{ public static void main(String[] a){ Box3D b=new Box3D(2,3,4); System.out.println(b.area()); Sy
```

7. Multiple Catch

```
class M{
    public static void main(String[] a){
        try{ int x=10/0; }
        catch(ArithmeticException e){ System.out.println("AE"); }
        catch(Exception e){ System.out.println("E"); }
    }
}
```

8. try-catch-finally

```
class M{
    public static void main(String[] a){
        try{ int x=5/0; }
        catch(Exception e){ System.out.println("Error"); }
        finally{ System.out.println("Finally"); }
    }
}
```

9. User Defined Exception

```
class MyEx extends Exception{}
class M{
    public static void main(String[] a)throws MyEx{
        int x=5;
        if(x<10) throw new MyEx();
    }
}
```

10. File Creation – OutputStream

```
import java.io.*;
class M{
    public static void main(String[] a)throws Exception{
        FileOutputStream f=new FileOutputStream("a.txt");
        f.write("Hello".getBytes());
        f.close();
    }
}
```

11. File Write – Writer

```
import java.io.*;
class M{
public static void main(String[] a)throws Exception{
    BufferedWriter w=new BufferedWriter(new FileWriter("d.txt"));
    w.write("Hi");
    w.close();
}}
```

12. Employee Pay Slip

```
class Emp{ String n; int id; double bp; }
class Prog extends Emp{
    void calc(){
        double da=0.97*bp,hra=0.1*bp,pf=0.12*bp,sc=0.001*bp;
        System.out.println("Net="+ (bp+da+hra-pf-sc));
    }
}
class M{ public static void main(String[] a){
    Prog p=new Prog(); p.bp=20000; p.calc();
}}
```

13. Prime Using Interface

```
interface P{ void check(int n); }
class M implements P{
    public void check(int n){
        int f=1; for(int i=2;i<n;i++) if(n%i==0) f=0;
        System.out.println(f==1?"Prime":"Not");
    }
    public static void main(String[] a){ new M().check(7); }
}
```

14. Inheritance Types

```
class A{}
class B extends A{}
class C extends B{}
class M{ public static void main(String[] a){ System.out.println("OK"); } }
```

16. JavaFX MenuBar

```
import javafx.application.*;
import javafx.scene.*;
import javafx.scene.control.*;
import javafx.stage.*;

public class M extends Application{
    public void start(Stage s){
        MenuBar mb=new MenuBar();
        mb.getMenus().add(new Menu("File"));
        s.setScene(new Scene(mb,200,100));
        s.show();
    }
    public static void main(String[] a){ launch(a); }
}
```

17. JavaFX Submenu

```
Menu f=new Menu("File");
Menu n=new Menu("New");
n.getItems().add(new MenuItem("Project"));
f.getItems().add(n);
```

18(i). Grid Layout

```
GridPane g=new GridPane();
g.add(new Button("B1"),0,0);
```

18(ii). Border Layout

```
BorderPane b=new BorderPane();
b.setTop(new Button("Top"));
```

19(i). Total & Average

```
class M{
    public static void main(String[] a){
        int m[]={80,70,60,90,85}, t=0;
        for(int x:m) t+=x;
        System.out.println("Total="+t+" Avg="+t/5);
    }
}
```

19(ii). Palindrome

```
class M{
    public static void main(String[] a){
        int n=121,rev=0,t=n;
        while(n>0){ rev=rev*10+n%10; n/=10; }
        System.out.println(t==rev?"Palindrome":"Not");
    }
}
```

20. Abstract Class Shape

```
abstract class Shape{ int a,b; abstract void printArea(); }
class Rect extends Shape{ void printArea(){ System.out.println(a*b); } }
class M{ public static void main(String[] a){ Rect r=new Rect(); r.a=2;r.b=3; r.printArea(); }}
```

21(a). Fibonacci

```
class M{
    public static void main(String[] a){
        int n=5,f=0,s=1;
        for(int i=0;i<n;i++){ System.out.print(f+" "); int t=f+s; f=s;s=t; }
    }
}
```

21(b). Factorial

```
class M{ public static void main(String[] a){
    int n=5,f=1; for(int i=1;i<=n;i++) f*=i; System.out.println(f);
}}
```

22. Inter-thread Communication

```
class Customer{
    int amt=10000;
    synchronized void withdraw(int x){
        while(amt<x){ try{wait();}catch(Exception e){} }
        amt-=x; System.out.println("Withdraw "+x);
    }
    synchronized void deposit(int x){
        amt+=x; System.out.println("Deposit "+x); notify();
    }
}
class M{
    public static void main(String[] a){
        Customer c=new Customer();
        new Thread()->c.withdraw(15000)).start();
        new Thread()->c.deposit(10000)).start();
    }
}
```